

إجابات اختبار مختبر الدوائر الكهربائية
الأسئلة النصية مع الخيارات والجواب الصحيح

1. The unit of current is:

- A. Ampere
- B. Volt
- C. Ohm
- D. Watt

A. Ampere: الجواب الصحيح

2. The current is measured by:

- A. Voltmeter
- B. Ammeter
- C. Ohmmeter
- D. Wattmeter

B. Ammeter: الجواب الصحيح

3. Extracting multiple (single-source) circuits from a multiple-sources circuit describes:

- A. Superposition method
- B. Thevenin method
- C. Norton method
- D. Nodal method

A. Superposition method: الجواب الصحيح

4. An ideal voltmeter has equivalent resistance:

- A. Zero
- B. Very small
- C. Infinity
- D. 1 ohm

C. Infinity: الجواب الصحيح

5. Node voltage method is based on:

- A. Kirchhoff Current Law (KCL)
- B. Kirchhoff Voltage Law (KVL)
- C. Ohm's law only
- D. Faraday's law

الجواب الصحيح: **A. Kirchhoff Current Law (KCL)**

6. Where does voltage division occur?

- A. Equal resistors only
- B. Parallel-connected resistors
- C. Series-connected resistors
- D. Both only when resistors are equal

الجواب الصحيح: **C. Series-connected resistors**

7. In the superposition method:

- A. Voltage sources become open and current sources short
- B. Both become open
- C. Voltage sources short-circuit and current sources open-circuit
- D. Both become short

الجواب الصحيح: **C. Voltage sources short-circuit and current sources open-circuit**

8. Where does current division occur?

- A. Series-connected resistors
- B. Equal resistors only
- C. Both series and parallel
- D. Parallel-connected resistors

الجواب الصحيح: **D. Parallel-connected resistors**

9. In a parallel connection:

- A. Source voltage equals the sum of resistor voltages
- B. Current through all resistors is the same
- C. Equivalent conductance equals the sum of conductances
- D. All answers are right

الجواب الصحيح: **C. Equivalent conductance equals the sum of conductances**

10. An ideal ammeter has equivalent resistance:

- A. Infinity
- B. 1 ohm
- C. Zero
- D. Very high

الجواب الصحيح: **C. Zero**

11. In a series connection:

- A. Equivalent resistance equals the sum of resistances
- B. Current is constant along the circuit
- C. Source voltage equals the sum of voltage drops
- D. All answers are right

الجواب الصحيح: **D. All answers are right**

12. The algebraic sum of voltages around any closed path is equal to:

- A. One
- B. Zero
- C. Infinity
- D. The source voltage

الجواب الصحيح: **B. Zero**

13. An ohmmeter is connected in:

A. Series

B. Parallel

C. Either without power

D. Short circuit

B. Parallel :الجواب الصحيح

14. Maximum power transfer occurs when:

A. $R_{load} = 0$

B. $R_{thevenin} = R_{load}$

C. $R_{thevenin} + R_{load} = 1$

D. R_{load} is infinite

B. $R_{thevenin} = R_{load}$:الجواب الصحيح

15. For three series resistors, voltage across R_1 is:

A. $V \times R_1 / (R_1 + R_2 + R_3)$

B. $V \times R_2 / (R_1 + R_2 + R_3)$

C. $V \times R_3 / (R_1 + R_2 + R_3)$

D. $V / (R_1 + R_2 + R_3)$

A. $V \times R_1 / (R_1 + R_2 + R_3)$:الجواب الصحيح

16. The resistance is measured by:

A. Ammeter

B. Voltmeter

C. Ohmmeter

D. Galvanometer

C. Ohmmeter :الجواب الصحيح

تنبيه: الأسئلة التي تعتمد على رسم دائرة أو قيم غير واضحة في الصورة تحتاج مراجعة الرسم الأصلي قبل اعتماد الإجابة.